

CatSniffer, a new open source hardware tool to sniff IoT signals

CatSniffer, a new portable, multi-protocol sniffer, supports multiple frequency bands common to IoT devices. Announced by open source design provider Electronic Cats, the new tool allows debugging, analysing and sniffing of IoT devices. The hardware is released under a CERN Open Hardware Licence v1.2.

CatSniffer covers three core frequency ranges grouped as LoRa, sub-GHz, and 2.4GHz. The protocols it covers include Thread, ZigBee, BLE 5, IEEE 802.15.4g, 6LoWPAN, LoRa (LoRaWAN), and sub-GHz.

The new sniffer relies on two radios – Semtech SX1262 handles LoRa and LoRaWAN communication to cover protocols. It was designed as a portable USB stick that integrates the new generation of the chips TI CC1352, Semtech SX1262, and Microchip SAMD21E17.

This board is an auditing tool for security researchers looking into IoT security. It can be used with different types of software including third-party sniffers such as SmartRF Packet Sniffer, Sniffle, zigbee2mqtt, Z-Stack-firmware, or custom firmware. Users can even write their own firmware for hacking needs.



“This kit is designed to allow product developers to evaluate electronic components, circuits, or software associated with the kit to determine whether to incorporate such items in a finished product, and software developers to write software applications for use with the end product,” read a company note.

The initial examples are available on the CatSniffer GitHub repo.

VLC 3.0.16, MPV 0.33.0, AbiWord 3.0.5, GIMP 2.10.24, Gnumeric 1.12.50, Chromium 90.0 and Mesa 21.0

Thales and IIIT-Delhi sign MoU on open source hardware R&D

Thales and IIIT-Delhi have signed an agreement for collaborative research and development in the field of open hardware and other allied subjects. The



collaboration will enable co-development of complex microprocessor architectures and multi-core processing systems using configurations available with the RISC-V open ISA specification. These systems will bring value to applications

in areas such as edge computing, smart manufacturing, defence and space.

The partners will collaboratively address safety issues in designing processors and embedded software for critical applications using the open source hardware approach. The activities have commenced from August 2021, and will be carried out in phases.

“We are delighted to join hands with an esteemed institution such as IIIT-Delhi. This partnership comes at an opportune time when technological advancements are fast shaping the world around us. It will draw on the expertise of Thales and scientific talent of IIIT-Delhi to enhance research and development in the area of open hardware and related subjects, and contribute to the rich innovation ecosystem in India,” said Ashish Saraf, VP and country director of Thales in India.

Prof. Ranjan Bose, director IIIT-Delhi, said, “This is a long-term strategic partnership and also a step towards strengthening Indo-French collaboration and cooperation.”

In India, Thales has formed partnerships with the Indian Institute of Science in Bengaluru, and the IITs in Chennai, Mumbai and Delhi.

Facebook open sources Winterfell, a STARK prover and verifier

Facebook has open sourced Winterfell, a library designed for developers to benefit from proofs of computational integrity (CI), which requires in-depth knowledge of cryptography to implement.

CI proofs are cryptographic technology that let developers do many tough things, including run a computation to get a result. Developers can use a CI proof to convince on computation correctness, regardless of the complexity or the need to rerun.

Winterfell is a general-purpose STARK prover and verifier that aims to bring CI to a wider audience by generating proofs for computations. The v0.1 version of the library is published on crates.io. It can generate CI proofs using STARK for any program that can be defined with Turing complete languages like Java, JavaScript or Perl. It is a STARK prover and verifier for arbitrary computations.

Facebook points out that there are still some technical challenges to overcome before proofs of computational integrity can be considered practical at a large scale.